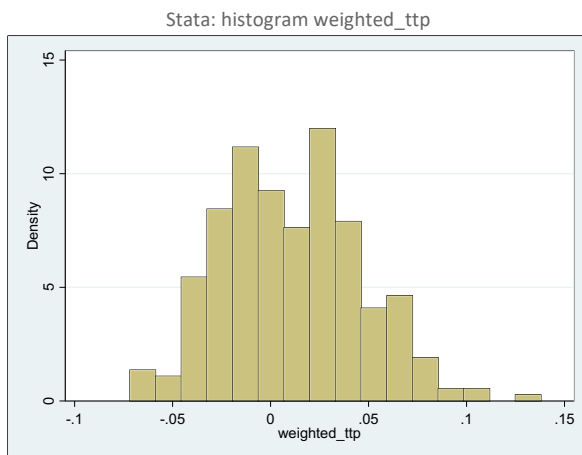


Decision 4 – Stata vs app: Research Baseline and Research Specification, 280 agents

Stata: Stata_File_Decision4_050926.dta and the figures in "Decision 4 - Rejected Transaction Defaults 040926-2". App: branch migration, commit d489143, browser screenshots, 7 Oct 2026.

Research Baseline – Options List Length – Tendency to Plan score

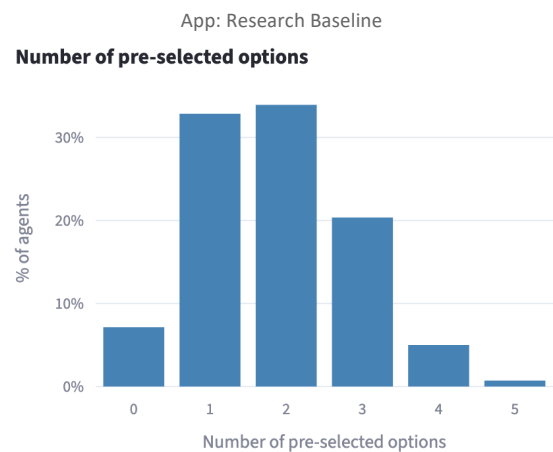
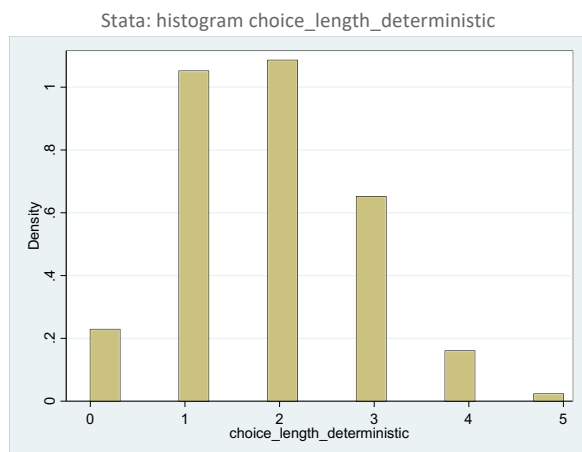
Agents 280 (participants in file order) · Income: not used · $\beta_0 = 0$ (Stata: no intercept); $-0.0152556564 \cdot zE + 0.0177638642 \cdot zA + 0.01959 \cdot zN + 0.00901465 \cdot zC + 0.0297 \cdot \text{education}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Baseline – Options List Length – number of pre-selected options

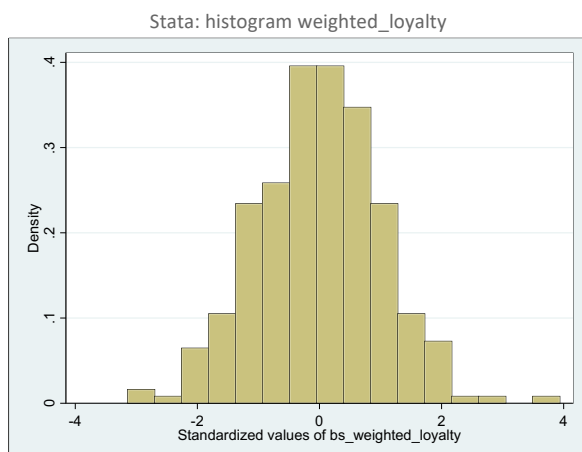
Agents 280 (participants in file order) · Income: not used · $\beta_0 = 0$ (Stata: no intercept); $-0.0152556564 \cdot zE + 0.0177638642 \cdot zA + 0.01959 \cdot zN + 0.00901465 \cdot zC + 0.0297 \cdot \text{education}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 6 bars with the same heights (7.1 / 32.9 / 33.9 / 20.4 / 5.0 / 0.7 % of agents for 0–5 options). Stata's y-axis is density, the app's is % of agents; the shape is the same.

Research Baseline – Loyalty – score

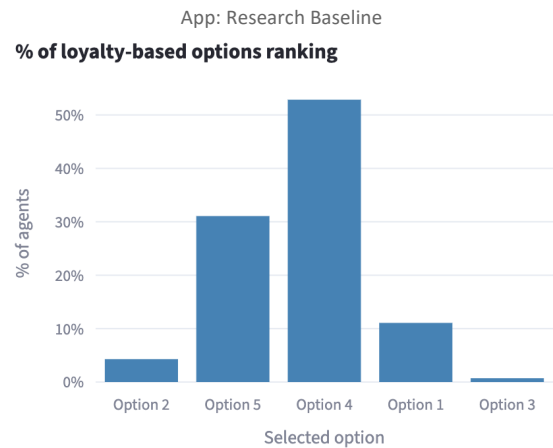
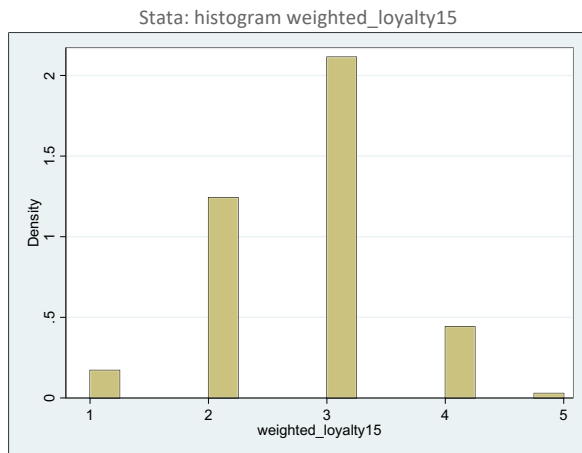
Agents 280 (participants in file order) · Income: not used · $\beta_1 = 0$ (Stata beta1 = 0); $-0.009828468 \cdot zE + 0.01096706 \cdot zO + 0.0123046 \cdot zA$, then standardised (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Baseline – Loyalty – options ranking

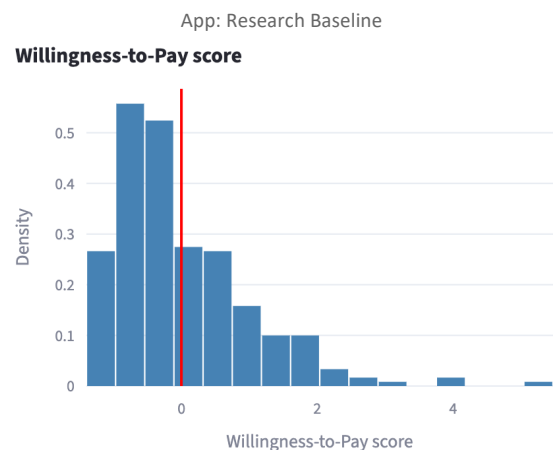
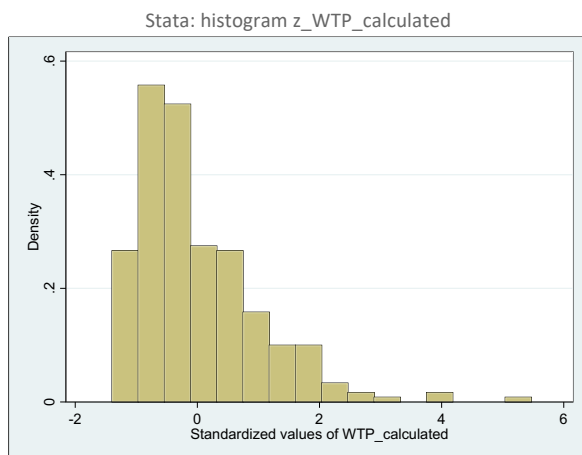
Agents 280 (participants in file order) · Income: not used · $\beta_1 = 0$ (Stata beta1 = 0); $-0.009828468 \cdot zE + 0.01096706 \cdot zO + 0.0123046 \cdot zA$, then standardised (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 5 bars in the same order and heights (4.3 / 31.1 / 52.9 / 11.1 / 0.7 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 2, 5, 4, 1, 3). Stata's y-axis is density, the app's is % of agents.

Research Baseline – Willingness-to-Pay – score (continuous income)

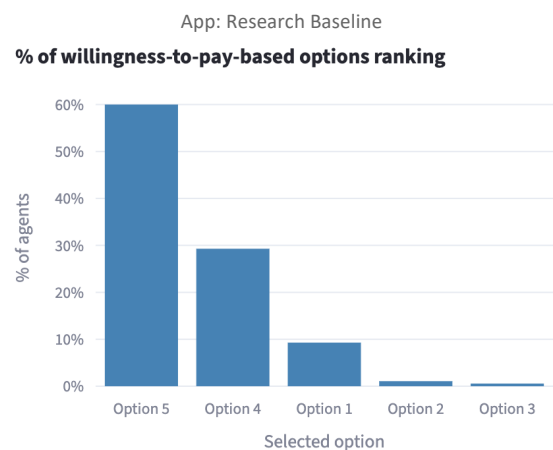
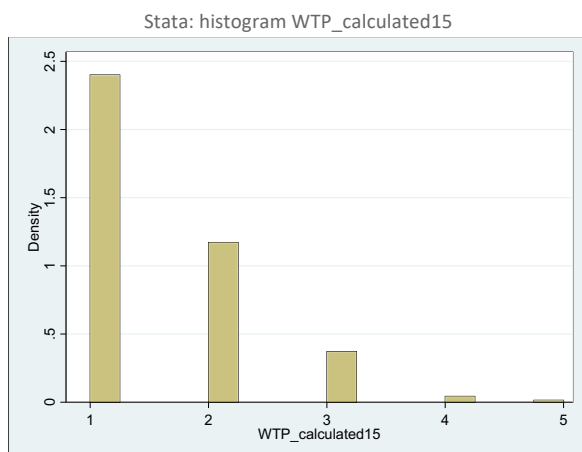
Agents 280 (participants in file order) · Income: Continuous only · $\beta_2 = 0$ (Stata beta2 = 0); $0.078863062 \cdot zE - 0.012326128 \cdot zA + 0.698 \cdot z \text{ income}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Baseline – Willingness-to-Pay – options ranking (continuous income)

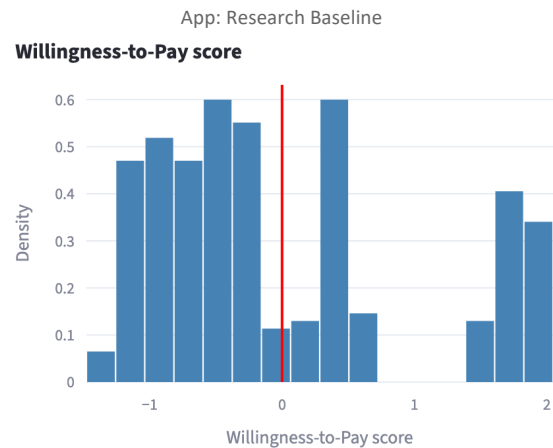
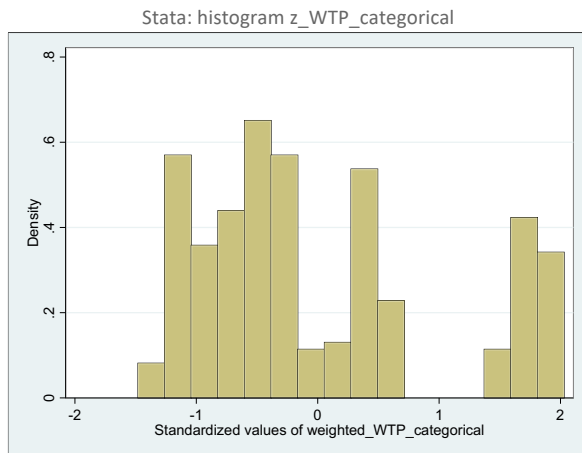
Agents 280 (participants in file order) · Income: Continuous only · $\beta_2 = 0$ (Stata beta2 = 0); $0.078863062 \cdot zE - 0.012326128 \cdot zA + 0.698 \cdot z \text{ income}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 5 bars in the same order and heights (60.0 / 29.3 / 9.3 / 1.1 / 0.4 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 5, 4, 1, 2, 3). Stata's y-axis is density, the app's is % of agents.

Research Baseline – Willingness-to-Pay – score (categorical income)

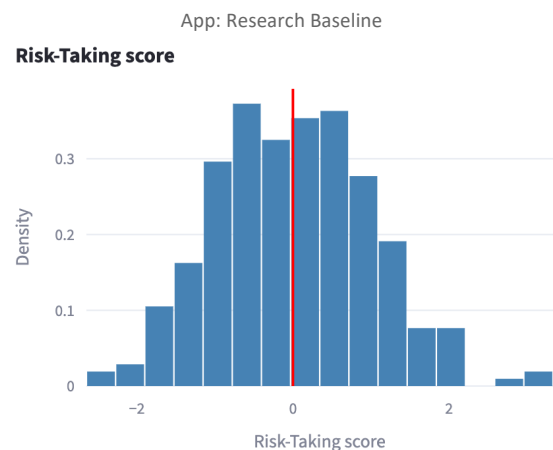
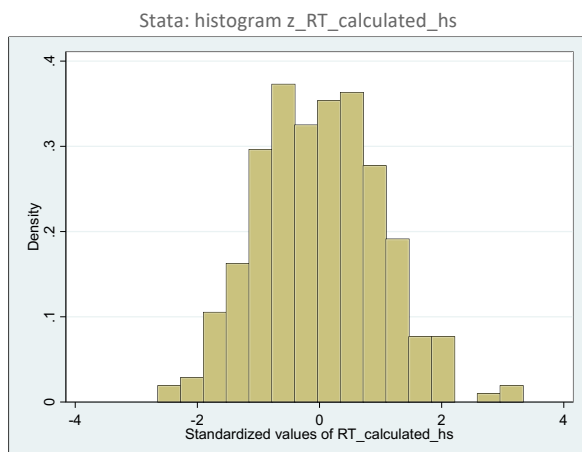
Agents 280 (participants in file order) · Income: Categorical only · $\beta_2 = 0$ (Stata beta2 = 0); allowance levels -0.691843 (base), $+0.2671588$, $+0.5057716$, $+0.9411747$, $+1.822471$; $0.078863062 \cdot zE - 0.012326128 \cdot zA$ (same as the Stata code in the document) · σ : none in Stata; Research Baseline never adds σ



Does not match – the Stata file's stored z_WTP_categorical was made with an older formula for the lowest allowance level (57 participants); the app uses the formula printed in the Decision 4 document and matches that formula bar by bar (max bar difference here 25% of the tallest bar).

Research Baseline – Risk-Taking – score (continuous income)

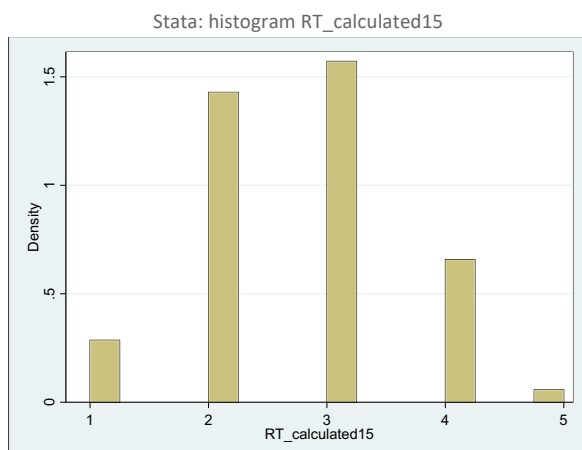
Agents 280 (participants in file order) · Income: Continuous only · $\beta_3 = 0$ (Stata beta3 = 0); $0.025942386297 \cdot zE + 0.023699214948 \cdot zO - 0.038734315188 \cdot zA - 0.037739440732 \cdot zC - 0.025388697852 \cdot zN + 0.006874197106 \cdot z \text{ income}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Baseline – Risk-Taking – options ranking (continuous income)

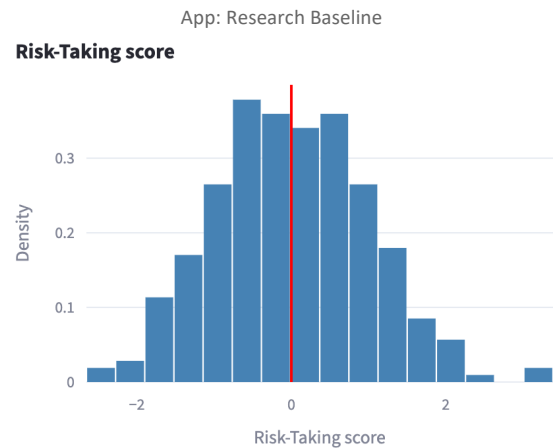
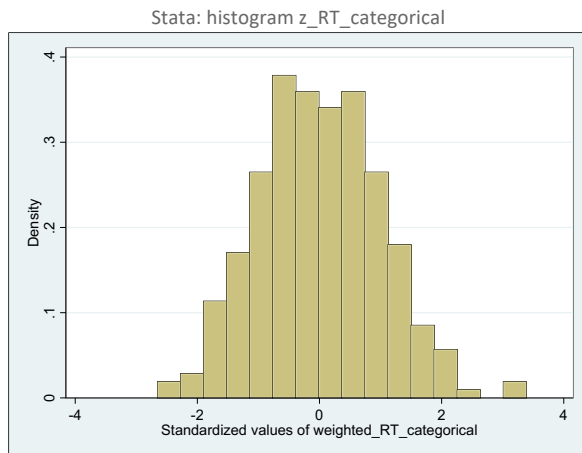
Agents 280 (participants in file order) · Income: Continuous only · $\beta_3 = 0$ (Stata beta3 = 0); $0.025942386297 \cdot zE + 0.023699214948 \cdot zO - 0.038734315188 \cdot zA - 0.037739440732 \cdot zC - 0.025388697852 \cdot zN + 0.006874197106 \cdot z \text{ income}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 5 bars in the same order and heights (7.1 / 35.7 / 39.3 / 16.4 / 1.4 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 5, 3, 1, 2, 4). Stata's y-axis is density, the app's is % of agents.

Research Baseline – Risk-Taking – score (categorical income)

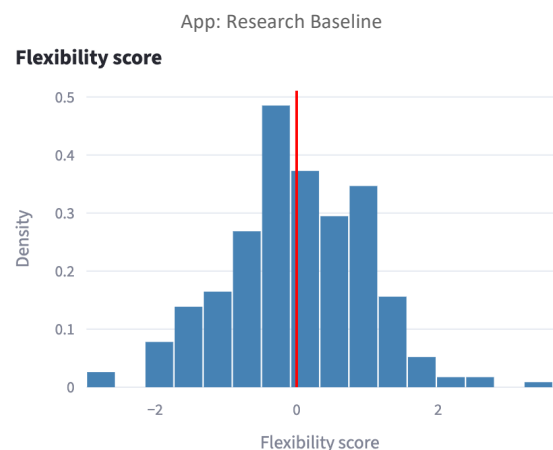
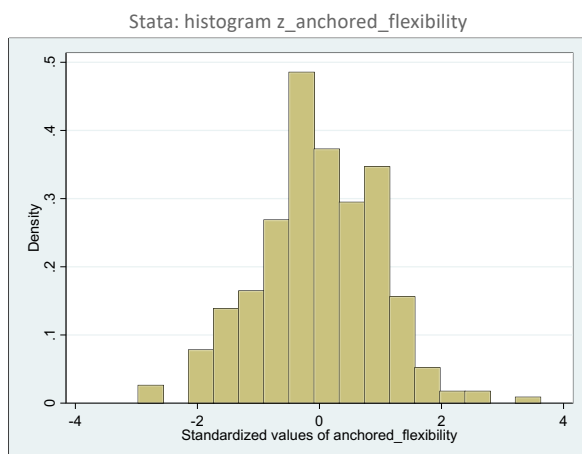
Agents 280 (participants in file order) · Income: Categorical only · $\beta_3 = 0$ (Stata beta3 = 0); allowance levels -0.0068307 (base), $+0.0026128$, $+0.0050555$, $+0.0092738$, $+0.0179812$; personality coefficients as in the continuous model (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Baseline – Flexibility – score

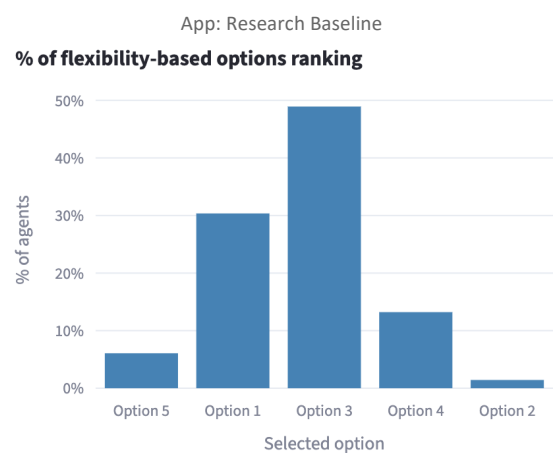
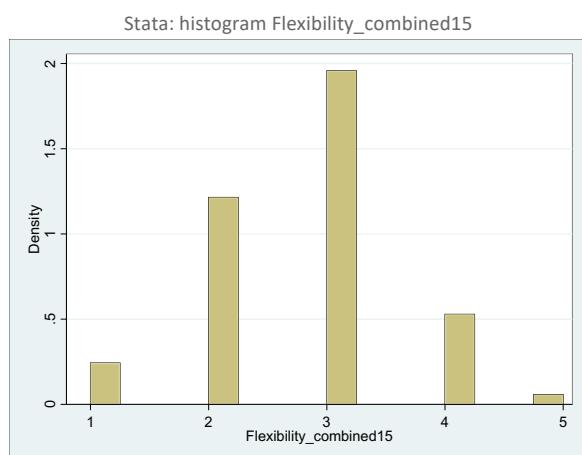
Agents 280 (participants in file order) · Income: not used · $\beta_4 = 0$ (Stata beta4 = 0); $0.0206 \cdot zE + 0.0294118 \cdot zO - 0.04921357 \cdot zN + 0.04339814 \cdot zA + 0.04811179 \cdot zC$, anchored $0.25 \cdot z$ stdactions + $0.75 \cdot \text{score}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Baseline – Flexibility – options ranking

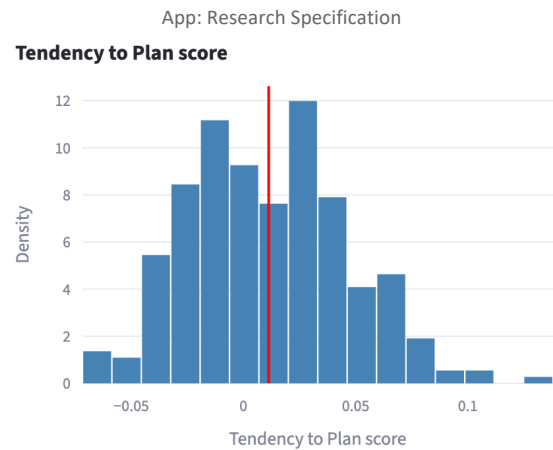
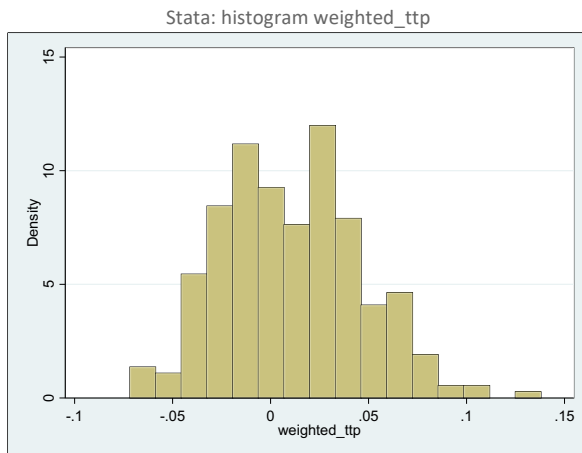
Agents 280 (participants in file order) · Income: not used · $\beta_4 = 0$ (Stata beta4 = 0); $0.0206 \cdot zE + 0.0294118 \cdot zO - 0.04921357 \cdot zN + 0.04339814 \cdot zA + 0.04811179 \cdot zC$, anchored $0.25 \cdot z$ stdactions + $0.75 \cdot \text{score}$ (same as Stata) · σ : none in Stata; Research Baseline never adds σ



Match – same 5 bars in the same order and heights (6.1 / 30.4 / 48.9 / 13.2 / 1.4 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 5, 1, 3, 4, 2). Stata's y-axis is density, the app's is % of agents.

Research Specification – Options List Length – Tendency to Plan score

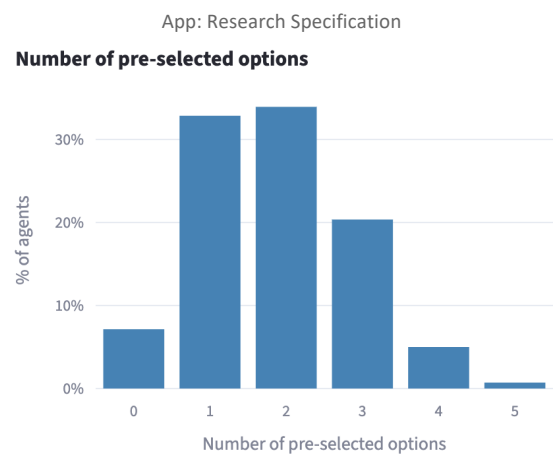
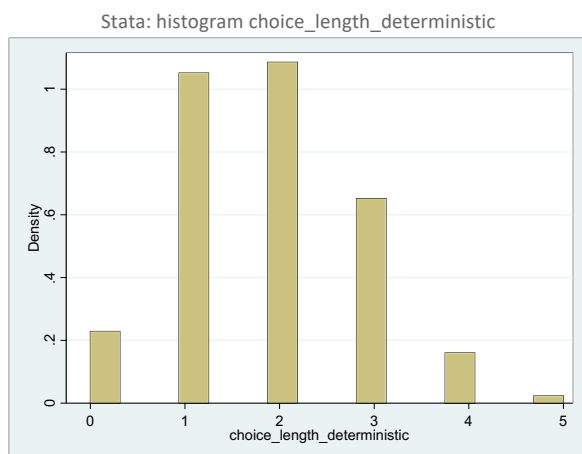
Agents 280 (participants in file order) · Income: not used · $\beta_0 = 0$ (Stata: no intercept); $-0.015255654 \cdot zE + 0.0177638642 \cdot zA + 0.01959 \cdot zN + 0.00901465 \cdot zC + 0.0297 \cdot \text{education}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Specification – Options List Length – number of pre-selected options

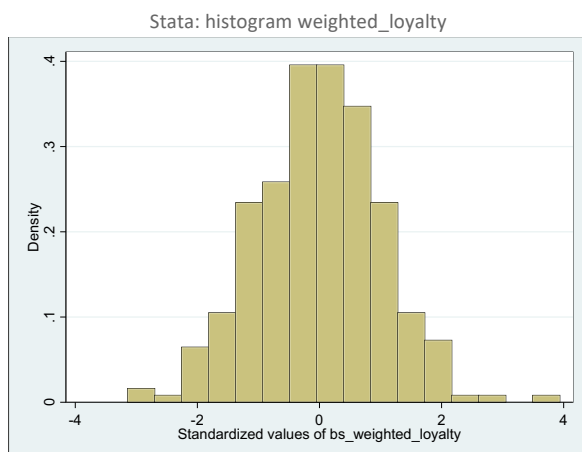
Agents 280 (participants in file order) · Income: not used · $\beta_0 = 0$ (Stata: no intercept); $-0.015255654 \cdot zE + 0.0177638642 \cdot zA + 0.01959 \cdot zN + 0.00901465 \cdot zC + 0.0297 \cdot \text{education}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 6 bars with the same heights (7.1 / 32.9 / 33.9 / 20.4 / 5.0 / 0.7 % of agents for 0–5 options). Stata's y-axis is density, the app's is % of agents; the shape is the same.

Research Specification – Loyalty – score

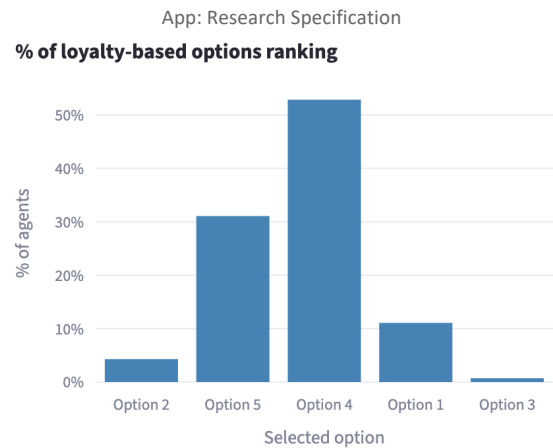
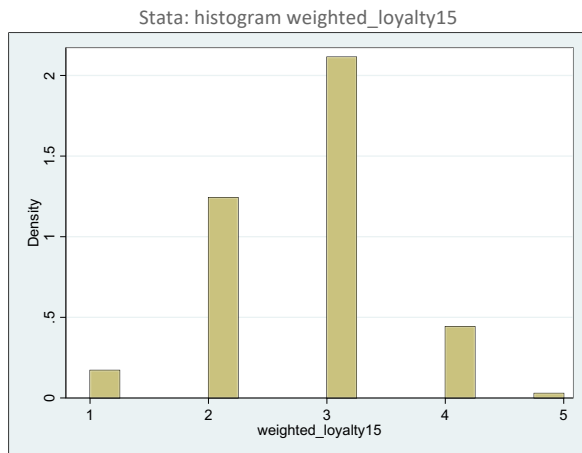
Agents 280 (participants in file order) · Income: not used · $\beta_1 = 0$ (Stata beta1 = 0); $-0.009828468 \cdot zE + 0.01096706 \cdot zO + 0.0123046 \cdot zA$, then standardised (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Specification – Loyalty – options ranking

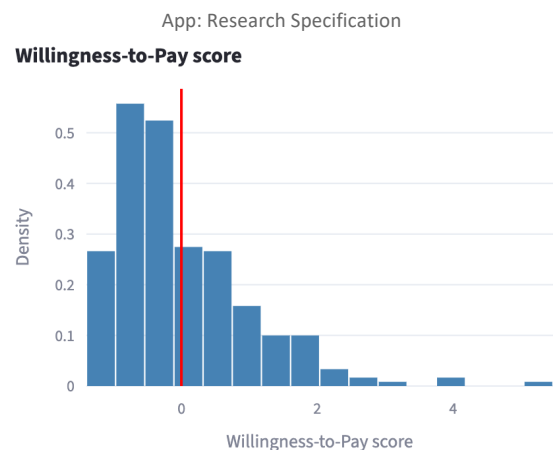
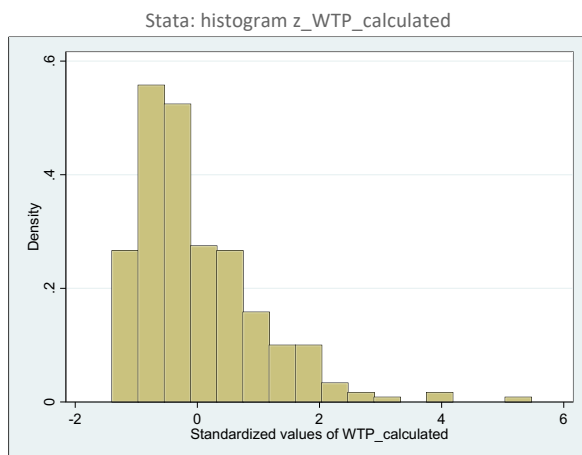
Agents 280 (participants in file order) · Income: not used · $\beta_1 = 0$ (Stata beta1 = 0); $-0.009828468 \cdot zE + 0.01096706 \cdot zO + 0.0123046 \cdot zA$, then standardised (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 5 bars in the same order and heights (4.3 / 31.1 / 52.9 / 11.1 / 0.7 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 2, 5, 4, 1, 3). Stata's y-axis is density, the app's is % of agents.

Research Specification – Willingness-to-Pay – score (continuous income)

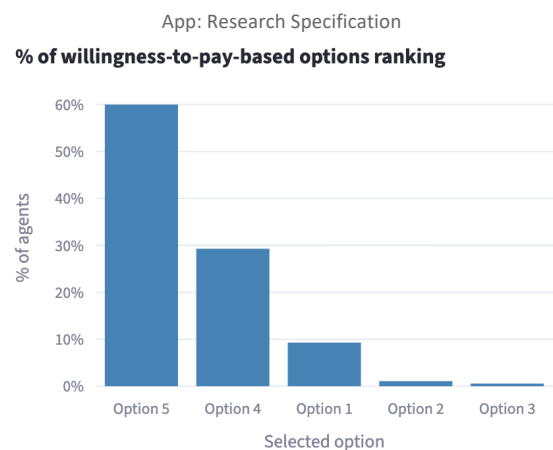
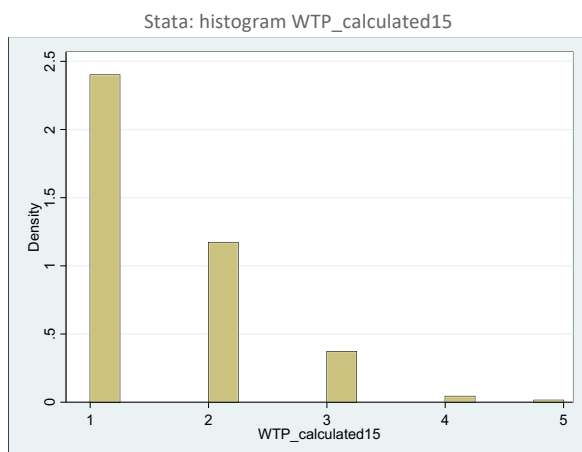
Agents 280 (participants in file order) · Income: Continuous only · $\beta_2 = 0$ (Stata beta2 = 0); $0.078863062 \cdot zE - 0.012326128 \cdot zA + 0.698 \cdot z \text{ income}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Specification – Willingness-to-Pay – options ranking (continuous income)

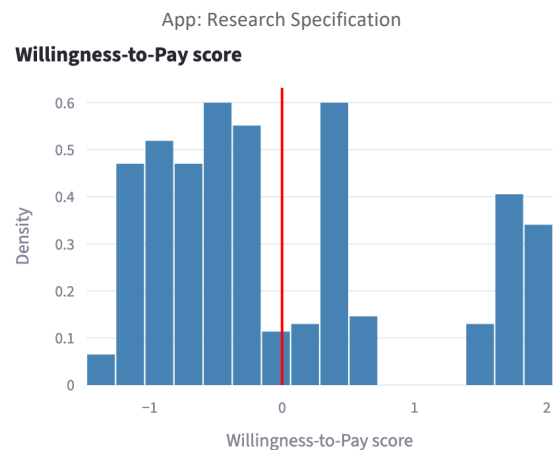
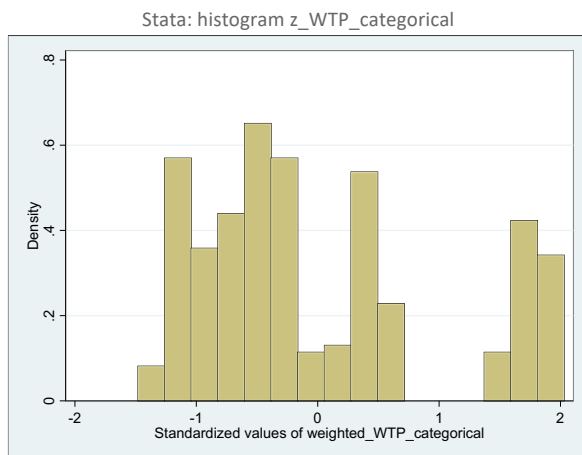
Agents 280 (participants in file order) · Income: Continuous only · $\beta_2 = 0$ (Stata beta2 = 0); $0.078863062 \cdot zE - 0.012326128 \cdot zA + 0.698 \cdot z \text{ income}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 5 bars in the same order and heights (60.0 / 29.3 / 9.3 / 1.1 / 0.4 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 5, 4, 1, 2, 3). Stata's y-axis is density, the app's is % of agents.

Research Specification – Willingness-to-Pay – score (categorical income)

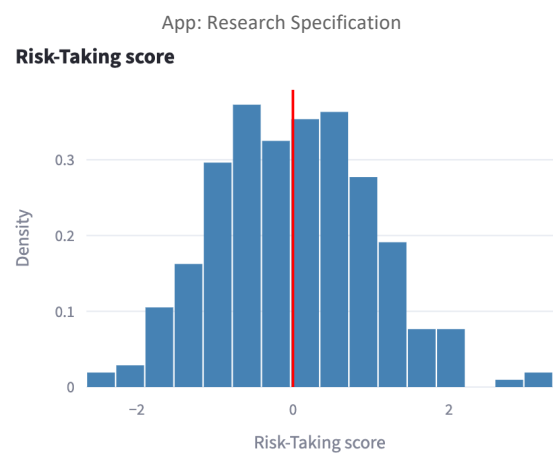
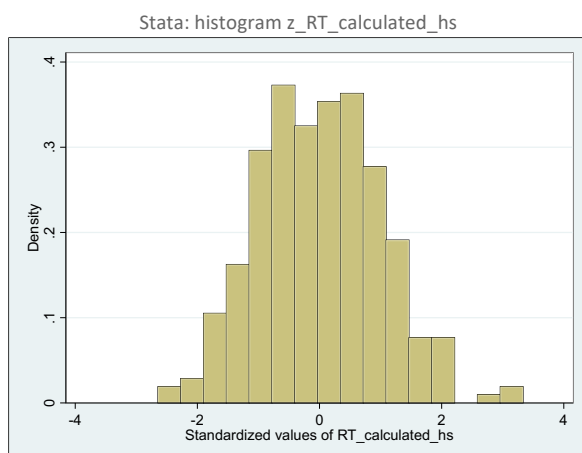
Agents 280 (participants in file order) · Income: Categorical only · $\beta_2 = 0$ (Stata beta2 = 0); allowance levels -0.691843 (base), $+0.2671588$, $+0.5057716$, $+0.9411747$, $+1.822471$; $0.078863062 \cdot z_E - 0.012326128 \cdot z_A$ (same as the Stata code in the document) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Does not match – the Stata file's stored z_WTP_categorical was made with an older formula for the lowest allowance level (57 participants); the app uses the formula printed in the Decision 4 document and matches that formula bar by bar (max bar difference here 25% of the tallest bar).

Research Specification – Risk-Taking – score (continuous income)

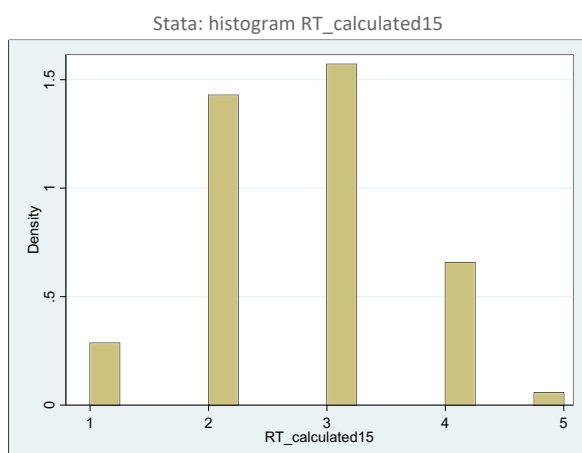
Agents 280 (participants in file order) · Income: Continuous only · $\beta_3 = 0$ (Stata beta3 = 0); $0.025942386297 \cdot z_E + 0.023699214948 \cdot z_O - 0.038734315188 \cdot z_A - 0.037739440732 \cdot z_C - 0.025388697852 \cdot z_N + 0.006874197106 \cdot z_{\text{income}}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Specification – Risk-Taking – options ranking (continuous income)

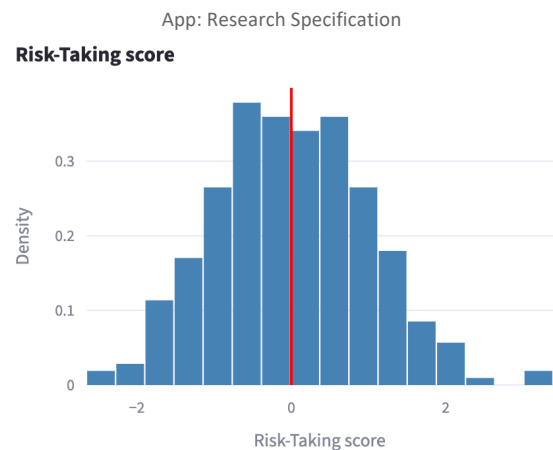
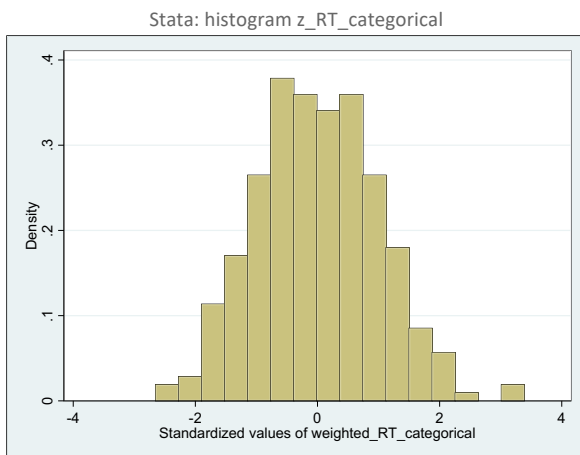
Agents 280 (participants in file order) · Income: Continuous only · $\beta_3 = 0$ (Stata beta3 = 0); $0.025942386297 \cdot z_E + 0.023699214948 \cdot z_O - 0.038734315188 \cdot z_A - 0.037739440732 \cdot z_C - 0.025388697852 \cdot z_N + 0.006874197106 \cdot z_{\text{income}}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 5 bars in the same order and heights (7.1 / 35.7 / 39.3 / 16.4 / 1.4 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 5, 3, 1, 2, 4). Stata's y-axis is density, the app's is % of agents.

Research Specification – Risk-Taking – score (categorical income)

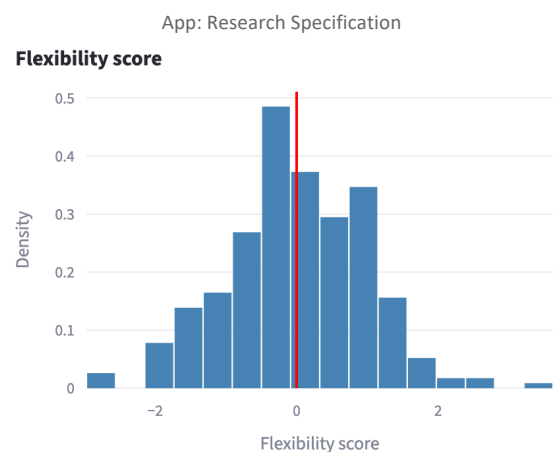
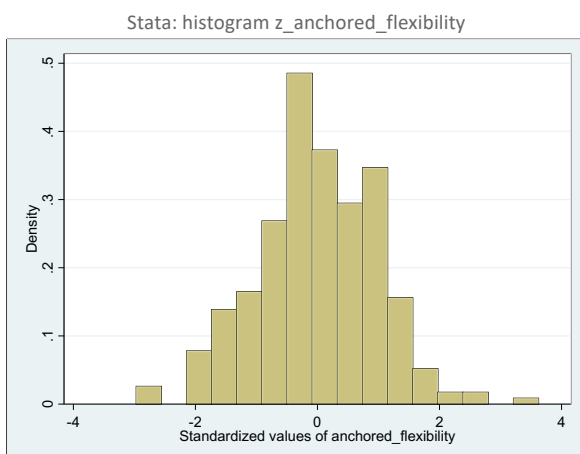
Agents 280 (participants in file order) · Income: Categorical only · $\beta_3 = 0$ (Stata beta3 = 0); allowance levels -0.0068307 (base), $+0.0026128$, $+0.0050555$, $+0.0092738$, $+0.0179812$; personality coefficients as in the continuous model (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Specification – Flexibility – score

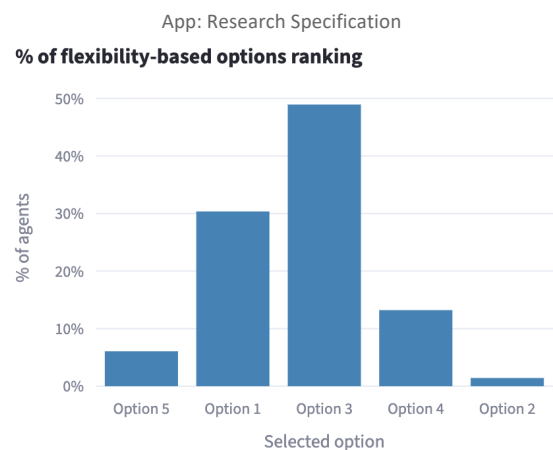
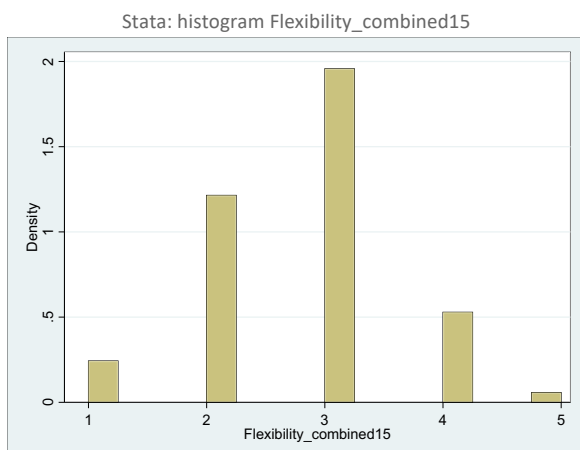
Agents 280 (participants in file order) · Income: not used · $\beta_4 = 0$ (Stata beta4 = 0); $0.0206 \cdot zE + 0.0294118 \cdot zO - 0.04921357 \cdot zN + 0.04339814 \cdot zA + 0.04811179 \cdot zC$, anchored $0.25 \cdot z$ stdactions + $0.75 \cdot \text{score}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 16 bins, same bin edges and bar heights (checked bar by bar against the Stata file). The red line in the app marks the mean (app only).

Research Specification – Flexibility – options ranking

Agents 280 (participants in file order) · Income: not used · $\beta_4 = 0$ (Stata beta4 = 0); $0.0206 \cdot zE + 0.0294118 \cdot zO - 0.04921357 \cdot zN + 0.04339814 \cdot zA + 0.04811179 \cdot zC$, anchored $0.25 \cdot z$ stdactions + $0.75 \cdot \text{score}$ (same as Stata) · σ : none in Stata, so σ is turned off (Normal(anchor, σ) draw unchecked)



Match – same 5 bars in the same order and heights (6.1 / 30.4 / 48.9 / 13.2 / 1.4 %). Stata numbers the segments 1–5; the app names each segment by its first option (Option 5, 1, 3, 4, 2). Stata's y-axis is density, the app's is % of agents.